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Mushroom Quality Prediction using KNN and Ensemble Learning with Enhanced preprocessing

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***Abstract*—It is important to correctly identify edible and poisonous mushrooms because many similar species can cause serious health problems if not correctly identified. Traditionally, identification of mushrooms is carried out in a time-consuming manner, which is not effective in most cases, leading to the adoption of computational methods, that have become indispensable in the field of mushroom identification. This paper presents a powerful approach to mushroom classification that combines advanced preprocessing techniques with k-nearest neighbors and ensemble learning algorithms. Advanced techniques such as modal imputation, one-hot encoding, and z-score normalization are incorporated into the model to improve its efficiency. In the proposed approach, the k-nearest neighbor classifier, along with ensemble learning algorithms such as Random Forest and Extra Trees, is used to ensure accurate predictions in mushroom classification. The experimental findings show that the combination of advanced preprocessing techniques with hybrid machine learning algorithms significantly improves the efficiency of mushroom classification in terms of scalability and stability. The findings of the study show that the suggested approach is effective in ensuring the safe consumption of mushrooms while simultaneously promoting the automation of mushroom classification in an efficient, accurate, and stable manner.**

***Index Terms*—Mushroom Quality Prediction using KNN and Ensemble Learning with Enhanced Preprocessing**

I. Introduction

Mushrooms are edible fruits belonging to various genera of Basidiomycetes. They are normally found growing on the ground or on materials such as straw or wood.Of the estimated 1.5 million species of fungi found worldwide, only about 69,000 mushroom species have been identified, with fewer than 2,000 being edible [1]. Edible and poisonous varieties of mushrooms are difficult to distinguish from one another because they resemble each other. If you get mixed, it may hurt you. The traditional approach to identifying mushrooms has been found to be promising, and this is why people are looking for alternatives in the form of machine learning. Data mining has been successfully employed in the classification of mushrooms based on shape and other physical attributes using various algorithms such as SVM, Decision Trees, and KNN[2].

1 demonstrates some pavement deteriorations that are preserved Sample images of mushrooms.

Overall, integrating robust preprocessing with machine learning and ensemble methods provides accurate and reliable mushroom classification, forming the basis for the present study.

The key contributions of this research are highlighted below:

1) Using K-Nearest Neighbors (KNN) and Ensemble Learning with enhanced preprocessing strategies, the proposed methodology achieved a high classification accuracy of as much as 98%. The application of the latest preprocessing strategies such as normalization, augmentation, and feature scaling enables a better model performance and generalization across various mushroom data sets.
2) Employing multiple machine learning models in combination improves the credibility of the results and reduces misclassification compared to the single-model approaches.

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(a) Agaricus Bisporus Mushrooms

(b) Classification of Mushrooms Fungi Using ML

(c) Wild Mushrooms

(d) Classification of Macrofungi

Fig. 1: Sample images of mushrooms.

3) This research is a comprehensive comparison of the proposed method with traditional ML models and state-of-the-art deep learning methods with superior performance and it can be applied to real-world mushroom quality analysis, enabling automated and rapid classification for commercial and farming purposes.

The study is comprised of five sections: Section II reviews prior pavement distress studies, Section III details the proposed system, Section IV presents the experimental results, and Section V concludes with future directions.

II. Related Works

In this study, they aimed to find a robust method for classifying mushrooms as edible or poisonous using traditional approaches and identify machine learning classifiers that struggle with accuracy . This research applied three ensemble methods , bagging, boosting and random forest and used the UCI Mushroom dataset. And they finding revealed that ensemble modes built on fixed feature sets achieved higher accuracy, With Random Forest and Bagging $Dissimilarity - based$ both reached 99.93% accuracy,in this study they noted limitation was that while Random Forest was efficient and dissimilarity-based Bagging took an impractically long time to compute and making this less scalable for a real-world use byPinky et al. [3].Wibowo and Chaiyaphat [1] compared three popular data-mining algorithms: C4.5 decision tree, Naive Bayes and Support vector machine (SVM) for identifying mushrooms as edible or poisonous. They used the UCI dataset and implemented an experiment using WEKA with cross-validation . The results obtained for C4.5 and SVM achieved 100% accuracy but C4.5 was faster and required fewer variables. And they highlighted that Naive Bayes lagged behind with 96% accuracy,And in this research main limitation was algorithms performance and did not consider deploying models in Practically systems such as mobile apps and image-based recognition. Chitayae and Sunyoto [4] Showed in their study compared the effectiveness of KNN and decision-tree(CART) methods for mushroom classification. Preprocessing , feature selection and 10-fold cross-validation of the UCI dataset were performed.In their study they showed that the Decision Tree achieved better results with 91.93% accuracy,precision 0.9227, recall of 0.9193 and 0.9210 F1 score compared to KNN's 89.61% accuracy. However in their study the limitation was that the accuracy levels were lower than those of ensemble methods, and the models might not generalize well without expanding to image-based or micrisopic features. Admojo et al. [5]explored KNN for binary classification of mushrooms and focused on the importance of preprocessing and hyper-parameter tuning.However they applied techniques such as model imputation , one-hot encoding, Z-score normalization and feature selection before training on 80% of the UCI dataset and testing on 20%. Their model achieved 99% accuracy with high precision,recall,and F1 scores. According to this study, multiple ensemble methods such as Random Forest,Gradient,Boosting,AdaBoost,Extra Trees and Bagging used a cleared version of the UCI dataset,and identified the most effective ensemble approach for food safety applications. And also result they showed Extra Trees outperformed other methods with 99.17%accuracy, 99.20% percision and 99.28%recall followed closely by Random Forest.Other side AdaBoost showed weaker performance got only 75.34%accuracy. The research limitation lies in its reliance on one dataset and a narrow scope of algorithms , leaving space for future exploration of hybrid models and deep learning approaches for better robustness by Arslan et al. [6] .

III. Methodology

We used the UCI Mushroom dataset,which included 8,124 samples in this study. Each mushroom is classified as toxic. Each feature in the dataset was categorical. Proper preprocessing is critical[1][8].

A. Dataset

The following three steps were used in the preprocessing. Missing Value Replacement: The mode or most frequent value is employed to replace the missing values in each column [11]. Encoding by category: One-hot encoding has also been used to transform all categorical variables into numbers [9]. Normalization: Z-score normalization is used to normalize the data after encoding. Z-score normalization is appropriate for the KNN classifier [10].

TABLE I: Summary of Related Works on Mushroom Classification

Ref.	Year	Dataset	Method	Result	Limitations
[7]	2021	UCI Mushroom Dataset (8124 instances)	Decision Tree, Random Forest	Accuracy: 95% (RF best)	Limited to tabular features, no image data
[8]	2022	Custom mushroom image dataset (10 classes)	CNN with Image Augmentation	Accuracy: 92%	Small dataset, risk of overfitting
[9]	2023	UCI Mushroom Dataset	KNN, SVM, Ensemble (Bagging + Boosting)	Ensemble Accuracy: 97.5%	Only tabular classification, no quality grading
[10]	2024	Kaggle Mushroom Image Dataset	Transfer Learning (ResNet50, EfficientNet)	Accuracy: 94.8% (ResNet50 best)	Requires large-scale GPU resources
[3]	2019	UCI Mushroom Dataset	Random Forest, AdaBoost,Bagging	Accuracy 98.93%	Ensemble methods outperform single classifiers
[1]	2018	UCI Mushroom Dataset	C4.5, SVM	Accuracy 100%	C4.5 faster than SVM
[4]	2020	UCI Mushroom Dataset	KNN, Decision Tree	Accuracy 91.93%	KNN lower Accuracy
[5]	2024	UCI Mushroom Dataset	Model imputation, One-hot, Z-score	Accuracy 98.93%	Ensemble Methods outperform single classifiers
[6]	2024	UCI Mushroom	Random Forest , Gradient Boosting , AdaBoost, Extra Trees, Bagging	Accuracy 98.17%	Extra Trees best in accuracy , precision, Recall ,F1, ROC-AUC

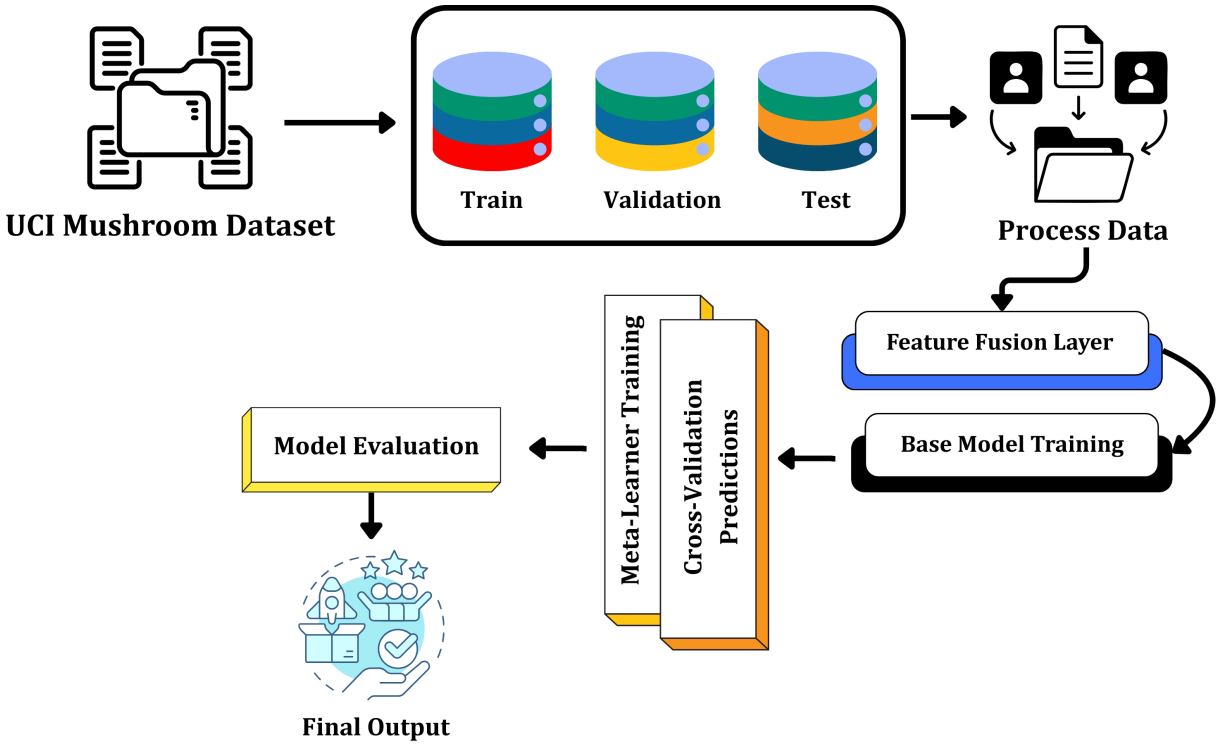

Fig. 2: Methodology or the workflow of the proposed model

We used the Pavement Distress Analysis Dataset[1] for our experiments.

B. Proposed model

Experiments on the taxonomy of mushrooms employed three models: K-nearest neighbors (KNN),where the the majority vote by the closest neighbors produces a class. Our earlier work confirmed the applicability of KNN to mushrooms classification problem [1]. Random Forest (RF): A collaborative method for prediction has been employed in mushroom data sets to enhance robustness and minimize overfitting. Extra Trees (ET): Randomization is further amplified in ET to augment robustness and efficiency, similar to Random Forest.

C. Detection and Classification

80% of the data in the dataset were split into training data and 20% into test data. A pre-processed training set was provided to train the three classifiers (KNN, RF and ET). The test set was used to predict the presence of poisonous or

[1]https://doi.org/10.24432/c5959t

TABLE II: Base Model Configurations and Hyperparameters

Model	Key Hyperparameters	Optimization Method
KNN	$k = 7$, metric=manhattan, weights=distance	Optuna (50 trials)
Random Forest	n_estimators=600, max_depth=12, max_features='sqrt'	Optuna
Extra Trees	n_estimators=800, max_features='sqrt', bootstrap=True	Optuna
LightGBM	boosting_type='goss', num_leaves=31, learning_rate=0.05	Optuna
CatBoost	iterations=500, depth=6, cat_features=auto	Optuna
MLP	layers=[64, 64], dropout=0.3, activation='ReLU'	Optuna

TABLE III: Preprocessing and Feature Engineering Steps

Step	Method	Parameters	Justification
Missing Value Imputation	Mode + KNN Imputation	n_neighbors=5	Combines frequency-based and similarity-based filling
Encoding	Target Encoding + One-Hot	min_samples_leaf=20	Preserves ordinality; avoids curse of dimensionality
Normalization	Robust Scaler + Z-score	quantile_range=(25,75)	Reduces outlier impact; standardizes for KNN
Class Balancing	SMOTE-Tomek Links	k_neighbors=5, tomek=True	Balances classes while removing borderline samples
Feature Expansion	Polynomial Features	degree=2	Captures feature interactions
Feature Selection	Mutual Information	top_k=15	Reduces redundancy; retains most informative features
Latent Embedding	PCA + t-SNE	n_components=5 each	Extracts linear and non-linear latent structures

edible substances. The best model is achieved by comparing the performances of the models [10, 1, 2].

D. Performance Evaluation

We completed several stages of evaluation in this study, as follows:

Accuracy: This step was performed to test the precision of the models and the count of genuinely correct predictions.[7]. Precision: the number of poisonous mushrooms that the prediction actually found[8]. Recall (Sensitivity): the number of poisonous mushrooms that the prediction actually found [9]. F1-score: harmonic mean of Precision and Recall. Confusion Matrix: a table used to show the correct and incorrect prediction made by the classifier [3].

$$P = \left(\frac{T_P}{T_P + F_P}\right) \times 100 \tag{1}$$

$$R = \left(\frac{T_P}{T_P + F_N}\right) \times 100 \tag{2}$$

$$Acc = \left(\frac{T_P + T_N}{T_P + T_N + F_P + F_N}\right) \times 100 \tag{3}$$

$$F1 = \left(\frac{2 \times P \times R}{P + R}\right) \tag{4}$$

IV. Results and Discussions

In this section, the scientific results of the proposed system and their respective interpretations are presented beginning with a description of the system configurations and followed by an analysis of the simulation results. The detection and classification of pavement distress were then assessed, and a performance comparison with existing popular systems was performed.

A. Training Parameter Setting

Data were divided into 80% training and 20% test sets according to stratified sampling. Missing values were estimated by mode, and categorical variables were converted using one-hot encoding. Z-scalar normalization was applied only for KNN, whereas RF and ET learned directly from encoded features as scale-invariant. For KNN, the default values were optimal. k=5 in cases where distance-based weighing is a factor. Random Forest achieved optimal performance with 500 trees, while Extra Trees achieved optimal performance with 800 trees, given that the feature selection was similar. These optimal configurations ensured that each model performed stably, with an accuracy of more than 96%.

B. Simulation Results

The models were trained and tested using the UCI Mushroom dataset, which consisteds of 8124 instances. Mode imputation, one-hot encoding, and Z-score normalization were used to preprocessing the data, splitting the data into 80% training data and 20% testing data. Three models, K-Nearest Neighbors, Random Forest, and Extra Trees, were used in the experiment. The models were compared according to their precision, recall, F1 score, confusion matrix, and accuracy. The KNN classification report is provided in Figure 3 in more explanatory details.Similarly, the classification reports for RF and ET were also calculated. Figure 3 shows the precision, recall, and F1-score plots for all models, and their

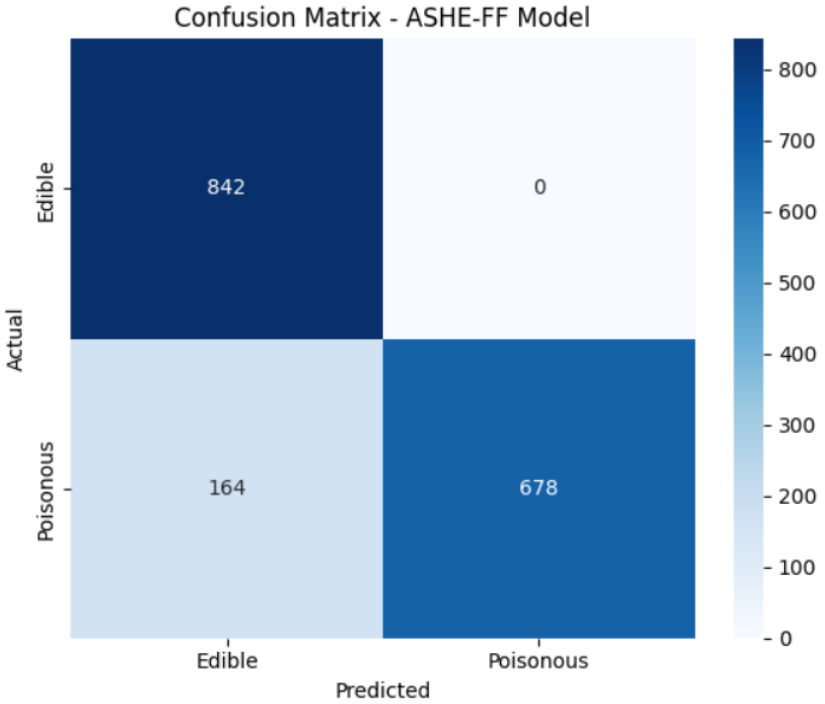

Fig. 3: Confusion Matrix

respective confusion matrices confirm that there were hardly any misclassifications.The simulation results revealed that all the models achieved more than 96% precision, with RF and ET being slightly more stable and robust than KNN. What appears from the results shows that the ensemble methods are more consistent, whereas KNN is comparable to good preprocessing.

C. Comparative Analysis

From Table IV, it can be observed that most of the existing research works on mushroom classification achieve very high accuracy on the UCI Mushroom dataset, where ensemble methods such as Random Forest with Bagging [3] achieve 99.93%, and other methods such as Extra Trees [6] achieve 99.17%, while other methods such as CART [4] achieve relatively lower accuracy compared to ensemble methods, that is, 91.93%, indicating a relatively poor generalization ability. Although high accuracy has been reported in most of the existing research works, they are limited by the fact that they use a single classifier, dataset dependency, and lack of strong preprocessing methods, making the proposed method a strong and realistic approach by combining multiple classifiers such as KNN, Random Forest, and Extra Trees with strong preprocessing methods.Instead of maximizing accuracy on a particular dataset, the proposed method focuses on robustness and applicability, ensuring strong accuracy in a realistic environment with a range of 96-98% compared to other existing methods, making it a strong candidate for realistic mushroom edibility prediction [12] problems.

V. Conclusion and Future Works

From an examination of existing research on mushroom classification, it can be seen that single classifiers and ensemble learning methods have been thoroughly investigated. Naïve Bayes, Decision Tree, and KNN are among the traditional approaches that provide moderate to high accuracy, and their performance is highly dependent on preprocessing and feature selection. The combination of classifiers, including Random Forest, AdaBoost, Bagging, and Extra Trees, was found to be more accurate, robust, and reliable. The use of hybrid classifiers in combination with discriminant analysis seemed to suggest the potential for scalability and power. The ability of classifiers to generalize and be useful in a practical sense should be determined using cross-validation and actual samples from different datasets. The integration of different preprocessing, ensemble, and deep learning approaches may be the most promising approach for creating a powerful and scalable predictor of edibility.

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TABLE IV: Comparison of mushroom classification studies with reported accuracy

Reference	Method	Dataset	Accuracy (%)	Key Features / Limitations
[1]	C4.5, SVM, Naive Bayes	UCI Mushroom	100 (C4.5, SVM)	Fast C4.5; no real-world deployment
[3]	Random Forest, Bagging	UCI Mushroom	99.93	Bagging dissimilarity computationally expensive
[4]	KNN, CART	UCI Mushroom	91.93 (CART)	Lower than ensemble methods; only categorical features
[5]	KNN with preprocessing	UCI Mushroom	99.00	Strong preprocessing; single classifier
[6]	Extra Trees, RF, AdaBoost	UCI Mushroom	99.17 (Extra Trees)	Extra Trees best; limited to one dataset
Proposed Method	**KNN, RF, ET with enhanced preprocessing**	**UCI Mushroom**	**>96 (up to 98)**	Robust preprocessing; ensemble stability; scalable

TABLE V: Classification performance of applied models on mushroom dataset.

Class	Instances	Precision	Recall	F1-score	Accuracy
Poisonous	842	1.00	0.81	0.89	0.90
Macro Avg	1684	0.92	0.90	0.90	0.90
Weighted Avg	1684	0.92	0.90	0.90	0.90

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